

Katie Byriel

EDUCATION

The University of Iowa // Iowa City, IA
Sustainability Science, BS
Geoinformatics - GIS, Certificate
Environmental Hydrological Science, Minor

Software Proficiencies:

Esri, Trimble, and Leica systems. ArcGIS Pro, Projectwise, Openroads, TopoDOT, Auto CAD, Civil 3D, Cyclone 3DR, Cyclone Register 360, Trimble RealWorks, GitHub, HTML, R-Studio, StoryMaps, Observable HQ, Survey123, Agisoft Metashape, Trimble Business Center, ENVI 5.7, QGIS, MaxEnt, Mapshaper, Excel

Geospatial Education:

- Intro to GIS
- Intro to Remote Sensing
- GIS: Environmental Applications
- GIS: Health and Epidemiological Applications
- Geographic Databases
- Geographic Visualizations
- LiDAR: Principles and Applications
- Geospatial Field Methods

Equipment Proficiencies:

EMLID Reach, Trimble DA2, Leica BLK360, Trimble X7 3D, GPS (R10, R12), Base Equipment, Total Stations (SX10, SX12, S6), Terrestrial Lidar, Drone Lidar/ Photogrammetry

WORK EXPERIENCE

Geospatial Survey Technician

Foth Engineering Solutions, Des Moines, IA

May 2025 - September 2026

- Civil Engineering
- AutoCAD, TopoDOT, Civil3D,
- Point Cloud processing with LIDAR data.
- Conducting survey support for municipal, transportation, and aviation related projects.
- Assisting with traditional survey activities that include utilizing digital level to set high accuracy elevations on project control and benchmark.
- Supporting operations using traditional land surveying tools (GPS/Total Stations) to complete topographic mapping surveys.
- Aiding construction staking efforts essential to accurate construction operations.
- Providing field support for 3D scanning and sUAS(Drone) operations.
- Learning basic survey fundamentals to complete topography surveys and legal boundary staking.
- Data extraction support associated with drone and mobile LIDAR scanning technology.

Production Stage Hand

Hancher Auditorium, Iowa City, IA

September 2025- May 2026

- Set up for performances; provide support during rehearsals, and sound checks.
- Working with various types of theatrical lighting, video, and sound equipment.
- Trained and tested on procedures, and in handling AV equipment.
- Providing security backstage and monitoring stage door access.
- Equipment set up, transport, and space restoration.
- Fire safety training.
- Approved to drive/operate university vehicles.

Ticketing Associate

University of Iowa Athletics, Iowa City, IA

August 2022-May 2026

- Working ticket sales for all sporting events throughout the academic year.

- Work with ticketing sales, and customer accounts.
- Handling customer ticket problems/ questions, and ticket reprints.
- Trained in credit card and money handling.
- Handling customer calls, preferences, complaints.

Premium Events Supervisor

University of Iowa Facilities, Iowa City, IA

August 2022- August 2025

- Work with event planning, sports management, ticketing sales, and customer accounts.
- Ran event set up, and commencement.
- In charge of gameday premium areas on behalf of the department and to provide support to other event staff.
- Responsible for event tear down, and security walk throughs.
- Control and setup computer systems and AV equipment.
- Strengthen communication, listening, and problem solving skills.

VOLUNTEER EXPERIENCE

UIowa Food Pantry, Iowa City, IA

August 2025-May 2026

- Shift Lead.
- Providing nutritional food and necessities to the Iowa City community.
- Helping to promote food security and accessibility.
- Gained insight into ways of helping others in my community.
- Knowledgeable on sustainable solutions for overconsumption.
- Improved communication, and leadership skills.

Friends of Hickory Hill Park, Iowa City, IA

April 2023 - May 2026

- Volunteer for maintaining sustainability in the park and controlling invasive species.
- Digging up invasive tree roots and garlic mustard.
- Climbing through timber to mark areas most taken over, and helped clean them up.
- Picking up litter throughout the park.
- Planting flowers for pollinators.

Saylorville Lake Restoration, Ankeny, IA

Summer 2022 & Summer 2023

- Worked to create sustainable solutions for waste disposal and wildlife restoration.
- Helped improve access to modes of waste disposal.
- Studied the patterns of bird and fish populations to better understand environmental needs.
- Helped determine causes for build up of toxic algae and propose solutions.

EXPERIENCE AND COLLEGE INVOLVEMENT

Iowa Community Integrated Geography Organization, ICIGO, Iowa City, IA

Spring 2024- May 2026

- Project Leader.
- Student GIS organization.
- Experience in an array of GIS Applications.
- Web content design, visualizations, and spatial analysis.
- Building of base maps, and project work in the Iowa City community.
- Project management and team management experience.
- Collaboration with non-profit organizations
- Use of LiDAR and adjacent scanning equipment.
- Knowledge using ArcGIS Pro, Arc Online, Cyclone, and Survey 123.

Projects:

- 3D Art Building — Point cloud extraction and animation.
- State of Iowa Disability Amenities — Mapping ADA access across the state.
- Cemetery Mapping - Burlington Risk Assessment
- Pilot Rock Scan
- Cedar County LESA Assessment
- LiDAR scanning: IMU, Theater Building, Art Exhibit, & Iowa City Riverbank.

Environmental Seminar, Iowa City, IA

Fall 2023

- Talked to professionals in a variety of environmental fields.
- Gained professional skills .
- Gained knowledge of varied careers such as land surveying, GIS, Research, America corps, and other leadership positions in the community.

Iowa Environmental Coalition, Iowa City, IA

Spring 2024- May 2026

- Student community involved in environmental related majors
- Attending informational lectures
- Volunteering within the community
- Planning bigger environmental events centered around conservation, and spreading education within the community.
- Forming connections with professionals in the community.